TEXORA · LECTURE SHEET

Chemical Finishing

Resins that crosslink inside cotton, silicones that soften and shed water, phosphorus chemistry that stops flame — the recipes, mechanisms and pad-dry-cure arithmetic of chemical finishing.

WP-4201 · TEXTILE FINISHING · TOPIC 3 · SPECIALIZATION — WET PROCESSING

1. How chemical finishes are applied

Almost every finish in this sheet reaches the fabric by one main route — **pad-dry-cure** — with exhaust, coating and foam as alternatives. The route must be understood first, because the recipe arithmetic belongs to it.

Pad-dry-cure line — the standard application route for resin and most functional finishes

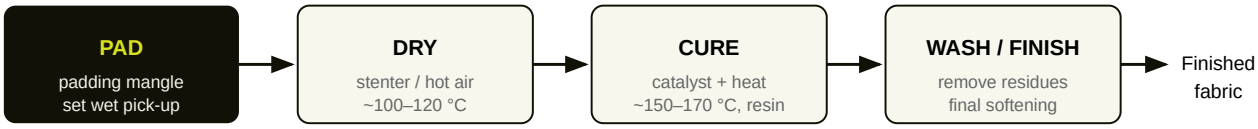


Figure 1 — Pad-dry-cure sequence: pad → dry → cure → wash-off/final treatment. Softeners and repellents may omit cure; resins cannot.

Wet pick-up is the mass of liquor carried by 1 kg of fabric after the padding nip, expressed as a percentage of dry fabric mass. It converts a bath concentration (g/L) into an amount actually delivered to the goods (g/kg, and then % owf).

Liquor carried (L/kg) ≈ wet pick-up (as a fraction) | Add-on (g/kg) = bath conc. (g/L) × liquor carried (L/kg) | % owf = add-on (g/kg) / 10

WORKED EXAMPLE 1 — PAD ADD-ON AT 70% WET PICK-UP

Wet pick-up 70% ⇒ each kilogram of fabric carries **0.70 L** of liquor.

DMDHEU resin at 60 g/L: add-on = 60 × 0.70 = **42 g/kg = 4.2% owf**.

Silicone softener at 20 g/L in the same bath: add-on = 20 × 0.70 = **14 g/kg = 1.4% owf**.

Lesson: g/L alone is not a dose. The same 60 g/L bath at 50% pick-up delivers only 30 g/kg — pick-up (nip pressure, fabric, speed) is part of the recipe.

METHOD	HOW IT WORKS	USED FOR
Pad-dry-cure	Continuous padding, drying, then high-temperature curing to react/fix the agent	Resins, repellents, flame retardants, soil release
Exhaust	Agent applied in a bath (jet, winch, garment machine) and exhausted onto the fibre like a dye	Cationic softeners, some silicones, antimicrobial and bio-polishing (batch)
Coating	A paste/film (PU, acrylic, PVC) spread on one face by knife or roller	Waterproofing, blackout, protective and technical fabrics
Foam application	Finish liquor whipped into foam, applied and collapsed on the fabric — very low water (see Topic 4)	Low-pick-up resin, repellent and softener application; energy saving

2. Resin / easy-care finishing

Cotton creases because hydrogen bonds between cellulose chains break under folding in the presence of moisture and re-form in the folded position. A **crosslinking resin** builds covalent bridges between chains inside the fibre, so the structure springs back towards flat — crease recovery, dimensional stability and wash-and-wear behaviour all rise together.

DMDHEU chemistry

The standard agent is **DMDHEU** (dimethylol dihydroxyethylene urea), an N-methylol compound. Under acid catalysis (commonly magnesium chloride, $MgCl_2$, often with a citric-acid co-catalyst) and heat in curing (about 150–170 °C for 1–3 minutes, or shorter at stenter-cure temperatures), its methylol ($–CH_2OH$) groups react with cellulose hydroxyls to form ether crosslinks. Low- and zero-formaldehyde variants (modified/etherified DMDHEU, and polycarboxylic-acid systems such as BTCA with sodium hypophosphite catalyst) reduce free formaldehyde at higher cost. Typical bath: DMDHEU 40–80 g/L, catalyst about 15–25% of the resin mass, softener and wetting agent as below, pH mildly acidic; pad at 65–75% pick-up, dry, cure, then a light wash-off where specified. Consequences to manage: crosslinking **reduces tensile and tear strength and abrasion resistance** (a 20–40% tear loss is common if over-treated), can release formaldehyde if poorly formulated or under-cured, and reduces dye affinity — so resin finishing always follows dyeing. Polyester-cotton needs less resin because the polyester component already recovers well.

> **ANTI-CREASE VERSUS WASH-AND-WEAR**
>
> **Anti-crease (crease-resist)** finishing targets recovery from creasing in wear — measured as crease recovery angle, mainly in the dry state, for fabrics that will still be pressed. **Wash-and-wear** finishing targets the whole care cycle: the garment must emerge from washing smooth, keep

its pleats/creases where wanted, and need little or no ironing — it is judged after washing (wet crease recovery and smoothness/appearance ratings) and needs higher, better-cured crosslinking, often with a wet-fixation or post-cure process. Every wash-and-wear fabric is anti-crease; an anti-crease fabric is not automatically wash-and-wear.

3. Softening finishes

SOFTENER CLASS	CHEMISTRY AND SUBSTANTIVITY	CHARACTER
Cationic	Quaternary ammonium compounds (e.g. distearyl dimethyl ammonium type); strongly substantive to negatively charged fibre surfaces, exhaust readily	Full, soft, slightly greasy luxury handle; can yellow whites, reduce light fastness of some dyes and block later repellent finishes
Silicone	Amino-functional polydimethylsiloxane micro-emulsions; spread as an ultra-thin lubricating film	Sleek, smooth, elastic handle with good sewability and tear-strength gain; hydrophilic silicone variants avoid water-absorbency loss
Non-ionic / anionic	Fatty-acid condensates, paraffin and polyethylene emulsions	Milder softness, safer for whites and for use inside resin baths; polyethylene also improves tear and abrasion performance

Softeners are combined with resin baths so one pad-dry-cure pass delivers easy care plus handle, and they partly compensate the harshness and strength loss of crosslinking. Over-softening is a real fault: seams slippage rises, absorbency of towels collapses, and silicone spots cause dye-resist marks if the fabric is reprocessed.

4. Water-repellent finishes

A repellent finish lowers the surface energy of each fibre so water beads and rolls off, while the fabric pores stay open and breathable — the opposite construction from a waterproof *coating*, which seals the pores.

TYPE	MECHANISM	DURABILITY AND LIMITS
Paraffin (wax) based, with aluminium/zirconium salts	Hydrocarbon film on fibres	Cheap, non-durable to semi-durable; poor oil repellency; stiffens handle
Silicone repellent	Polysiloxane film, cured with catalyst	Good water repellency, soft handle, semi-durable to durable; no oil repellency; can

		reduce soil release
Fluorocarbon (fluoropolymer)	Perfluoroalkyl side chains present a very low-energy CF_3 surface	Water *and* oil repellency, the most durable class; highest cost; modern products use short-chain (C6) chemistry under environmental regulation

Testing is by spray rating (AATCC 22 / ISO 4920) before and after washing, and by hydrostatic head only where a waterproof claim is made. Repellency fails early if the fabric carries residual surfactant or cationic softener — cleanliness before padding is part of the finish.

5. Flame-retardant finishes

Cellulosics burn readily; a flame-retardant (FR) finish intervenes by promoting char formation and starving the flame of fuel gases, or by releasing flame-quenching species. Cotton FR is a certified, safety-critical finish — recipe control and wash durability decide legal compliance.

CLASS / PRODUCT TYPE	MECHANISM	DURABILITY
Non-durable: borax–boric acid, ammonium phosphates/sulfamate salts	Char promotion and glassy coating; water-soluble salts	Removed by washing (and by rain); for curtains and disposables not laundered
THPC system (Proban type)	Tetrakis(hydroxymethyl)phosphonium chloride condensed with urea, padded, ammonia-cured to an insoluble polymer inside the fibre, then oxidised	Durable to 50+ washes; soft handle retained; process needs an ammonia chamber
Pyrovatex type (N-methylol phosphonopropionamide)	Reactive phosphorus compound co-cured with the resin system by pad-dry-cure, binding P to cellulose through methylol chemistry	Durable; simpler pad-dry-cure line, but handle stiffer and strength loss must be controlled

Phosphorus–nitrogen synergy is the working principle in both durable systems: phosphorus drives char, nitrogen compounds assist fixation and char stability. FR fabrics are tested to the end-use standard (for example EN ISO 11612 for protective clothing, BS 5867 for curtains) *after* the specified wash programme — durability is the certificate, not the fresh finish.

6. Soil-release, antistatic and antimicrobial finishes

- **Soil release:** hydrophilic polymer finishes (polyethylene-glycol / acrylic or fluorochemical hybrid types) on polyester and blends. The fibre surface becomes water-loving in the wash, so oily soil that clings to untreated polyester is lifted and rinsed away, and redeposition is reduced. Judged by stain-release ratings after washing, not by how clean the fabric looks when new.
- **Antistatic:** static charge builds on dry synthetics, causing cling, dust attraction and sparks. Hygroscopic antistats (quaternary and ethoxylated types) hold a moisture film that leaks charge away — effective but humidity-dependent and semi-durable; durable routes use conductive fibres/yarns or inherently dissipative polymers in technical textiles.
- **Antimicrobial:** agents that kill or inhibit bacteria and fungi on the fabric — silver-ion systems, quaternary silane (Si-Quat) types that bond to the surface, PHMB and triclosan-free organic systems. Used in socks, sportswear, medical and hospital textiles. Claims must name the test (e.g. AATCC 100, ISO 20743) and the number of washes survived; odour control is the usual consumer-level goal.

7. Enzyme / bio-polishing as bio-chemical finishing

Cellulase enzymes hydrolyse the loose microfibrils (fuzz) protruding from cotton yarn surfaces. In **bio-polishing** (typically acid cellulase, pH about 4.5–5.5, 50–55 °C, batch exhaust, followed by enzyme deactivation at high temperature or alkaline pH), the fuzz is weakened and removed by mechanical action in the same bath. Result: a cleaner, smoother surface, better print definition, less pilling and a softer appearance that survives washing — because fibres are physically removed, not coated. Weight and strength fall slightly (a few percent) and must be allowed for. Bio-polishing sits between dyeing stages or after dyeing depending on shade control, and it is the model of a bio-chemical finish: a biological catalyst doing a finishing job with low chemical load.

8. Choosing and controlling a chemical finish

Control points common to all recipes: bath pH and temperature stability; pick-up verified by weighing dry and padded fabric; drying without migration (agent carried to one face by uneven drying shows as two-sided effects); curing temperature profile measured across the width; and retained samples washed and re-tested for the claimed property. Compatibility in a combined bath is checked in the laboratory first — cationic softener plus anionic repellent in one bath, for instance, will precipitate and ruin both.

9. Glossary — key terms

- **Crosslinking:** covalent bridges between polymer chains inside a fibre, created by a curing resin.
- **DMDHEU:** the standard N-methylol resin for durable-press/easy-care finishing of cellulosics.
- **Wet pick-up:** liquor carried after padding, as % of dry fabric mass; converts g/L into add-on.

- **Add-on:** agent actually deposited, in g/kg or % owf.
- **Curing:** the heated reaction step that fixes a finish (crosslinking, polymerisation, film formation).
- **Soil release:** finish property that lets oily soil wash out of hydrophobic fibres.
- **Bio-polishing:** cellulase removal of surface fuzz from cellulosic fabrics.

10. Key points

- Add-on = g/L × pick-up fraction: DMDHEU 60 g/L at 70% gives 42 g/kg (4.2% owf); silicone 20 g/L gives 14 g/kg.
- DMDHEU + $MgCl_2$ catalyst + cure crosslinks cellulose — easy care rises, strength falls; manage both.
- Repellency keeps pores open; coating closes them. Fluorocarbon repels oil as well as water.
- Durable cotton FR is phosphorus chemistry fixed inside the fibre (THPC, Pyrovatex); salts are non-durable.
- Wash-and-wear is judged after washing; anti-crease mainly on crease recovery in wear.

11. Summary

Chemical finishing writes new properties into the fabric: crosslinked cellulose that springs flat, silicone films that soften, low-energy surfaces that shed water and oil, phosphorus locked in the fibre that chars instead of flaming, hydrophilic surfaces that release soil, charge-leaking and microbe-controlling treatments, and enzymes that polish the surface biologically. All of it is dosed through one piece of arithmetic — bath concentration times wet pick-up — and fixed by drying and curing discipline on the line drawn in Figure 1. Topic 4 now catalogues the machines and agents themselves.

12. Practice

1. A pad bath contains DMDHEU 80 g/L and wet pick-up is 65%. Calculate the resin add-on in g/kg and % owf.
2. A fluorocarbon repellent is padded at 30 g/L with 70% pick-up. Find the add-on, and state two properties it gives that a paraffin repellent does not.
3. Explain why a resin-finished cotton loses tear strength, and name two recipe measures that limit the loss.
4. A hotel wants curtains that will not be washed, and workwear washed weekly for a year. Choose an FR class for each and justify the choice.
5. Why is bio-polishing described as a durable finish even though nothing is added to the fabric?

Answer key

1. Add-on = 80 × 0.65 = 52 g/kg = 5.2% owf.
2. Add-on = 30 × 0.70 = 21 g/kg = 2.1% owf. Fluorocarbon additionally repels oil and is durable to repeated washing; paraffin repels water only and is non- to semi-durable (either two).
3. Crosslinks lock cellulose chains so they cannot slip and share load at a tear tip; stress concentrates and chains break together. Limit by using the lowest resin level that meets the crease target and by adding a polyethylene/silicone softener (lubrication lets yarns move) — also correct curing without over-baking (any two sound measures).
4. Curtains: non-durable borax–boric acid or ammonium salt FR is acceptable because there is no laundering to remove it (verify the end-use standard permits it). Workwear: a durable system — THPC (Proban type) or Pyrovatex type — certified after the specified number of washes.
5. Because the change is a physical removal of surface microfibrils. There is no coating to wash off; the polished surface, lower pilling and clearer appearance persist for the garment's life (with only slight weight/strength cost).

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